



Chapter I

Introduction

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CONTENT

1. Overview
2. Components of a Web-based System
3. HTTP Protocol
4. URL
5. HyperText & HyperLink
6. Web Cache



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1. OVERVIEW

Web?

World Wide Web, www, web, w3 is an Internet service

Is the most popular and convenient form of information organization



- Allows searching for information resources through hypertext using hyperlinks.
- These are text documents usually stored with extensions such as .html, .htm, etc.
- The Web was invented and put into use around 1990 by British academic Tim Berners-Lee at CERN. Since then, the web has developed rapidly and become a hypermedia system.

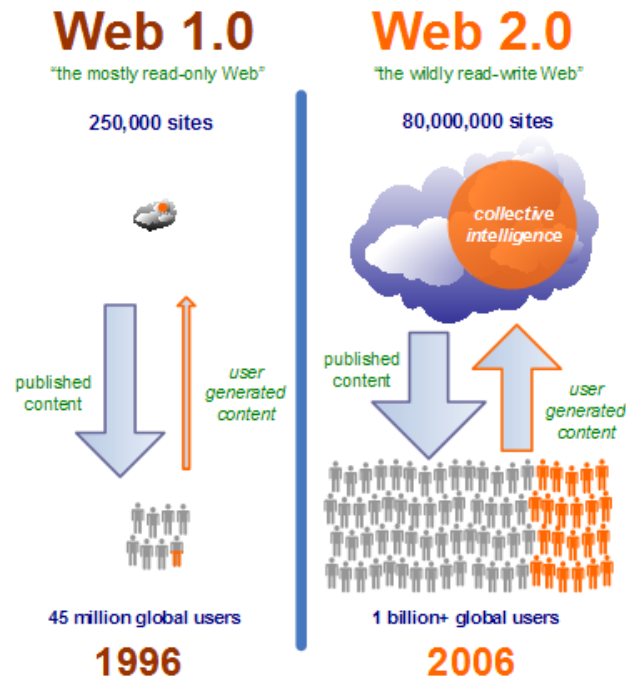
◆ WEB GENERATIONS

Web 1.0 (Static web):

- Information is one-sided
- Websites are mainly owned by companies, organizations, and news agencies.
- HTML
- Lack of interactivity

Web 2.0 (Dynamic web):

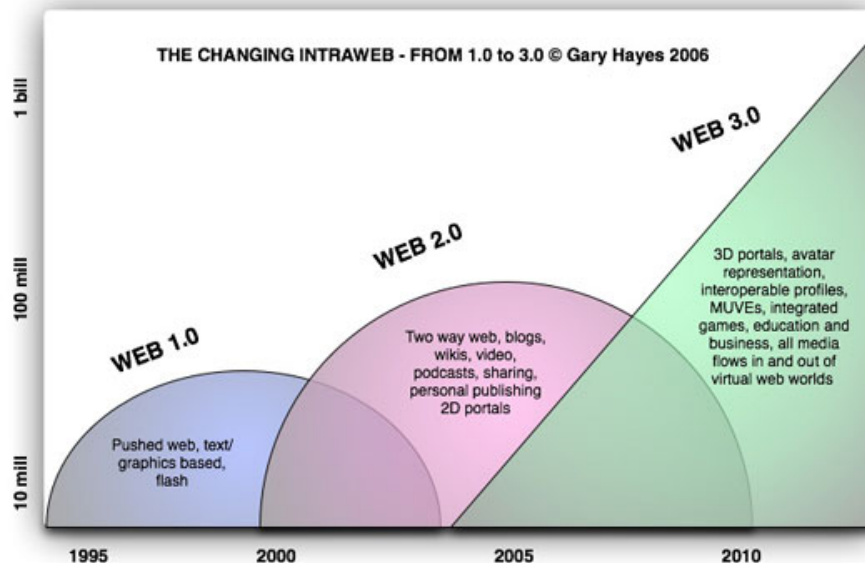
- Information is bidirectional
- It is community-oriented
- Server-client technology, content delivery mechanisms, and communication
- The web serves as a platform, capable of running any application
- It can run on multiple devices with a rich application interface



WEB GENERATIONS

Web 3.0 (Web of Data): More Intelligent Web

- An extension of Web 2.0
- Semantic web applications
- Real-time web applications



WEB DEVELOPMENT

What are the different types of web development?

**Frontend
Development**



**Backend
Development**



**Full Stack
Development**

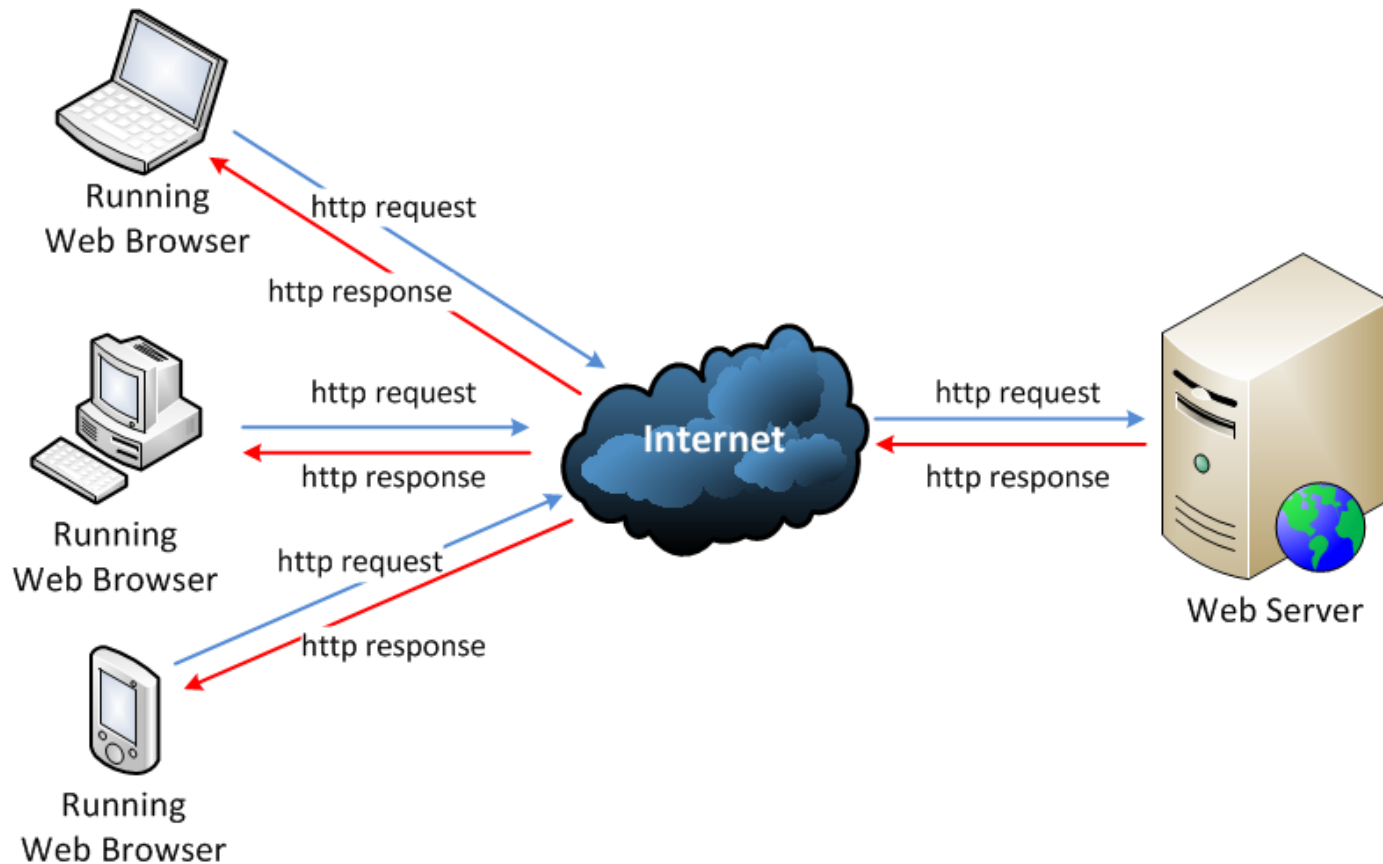




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2. COMPONENTS OF A WEB-BASED SYSTEM

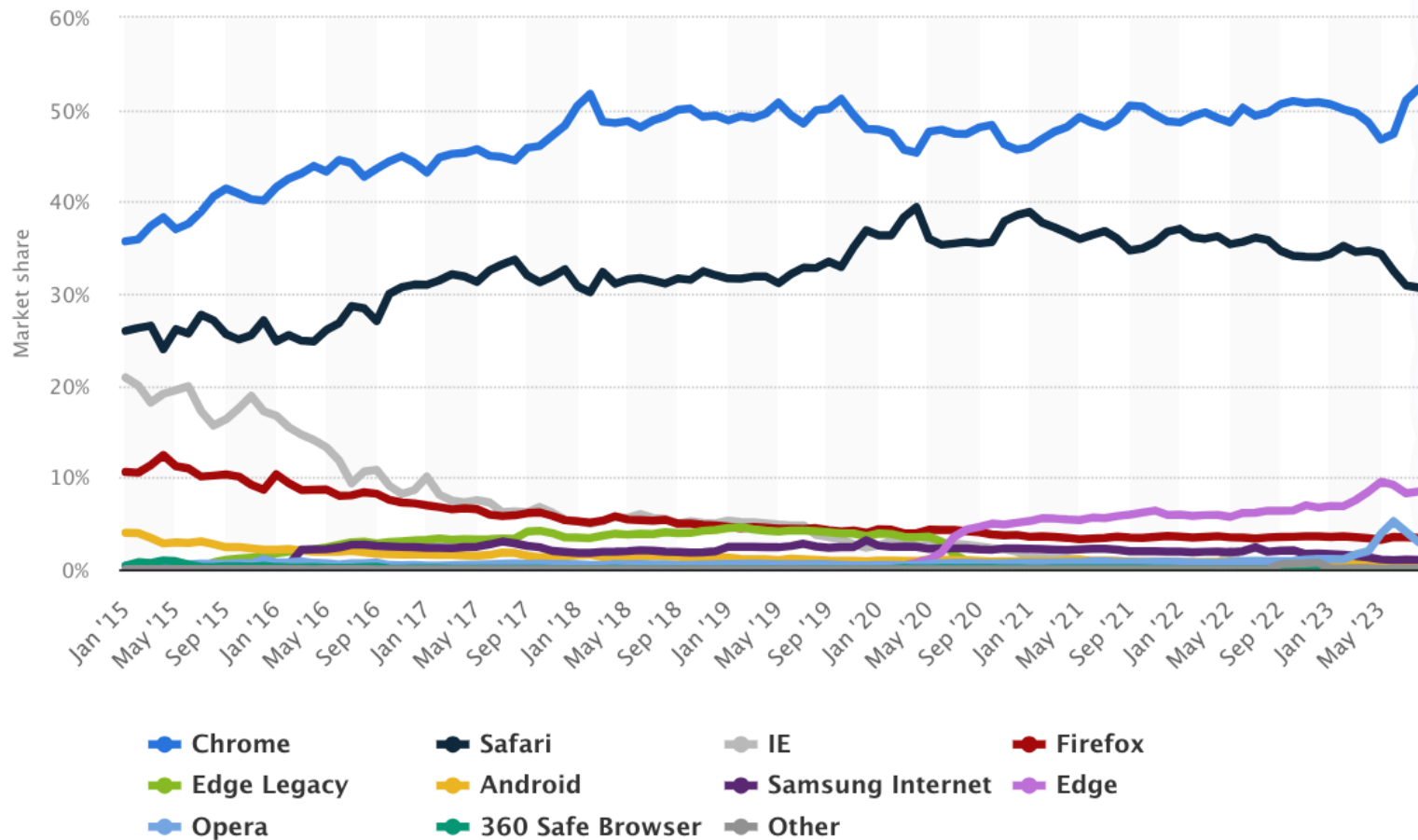


◆ Web client

Web Browser: Internet Explorer, Firefox, Chrome, Safari, Opera, Netscape, Mozilla, ...

- Installed on the client machine.
- Sends web requests to the Web Server.
- Receives results from the Web Server.
- Displays the results.

Web client



Source: statista.com

Web Browser

Addressing the diversity of web browsers:

- Don't invest in minor differences
- Follow common standards
- Use HTML with correct syntax, precise and clear structure
- Avoid using HTML elements that are specially supported by only one or two browsers

◆ Web server

Web Server: Apache, Tomcat, NGINX, MS Internet Information Server, lighttpd, ...

- Installed on a server, providing a Web service.
- Listens for Web requests on a port (80).
- Processes the requests.
- Generates results and returns them to the Web browser.



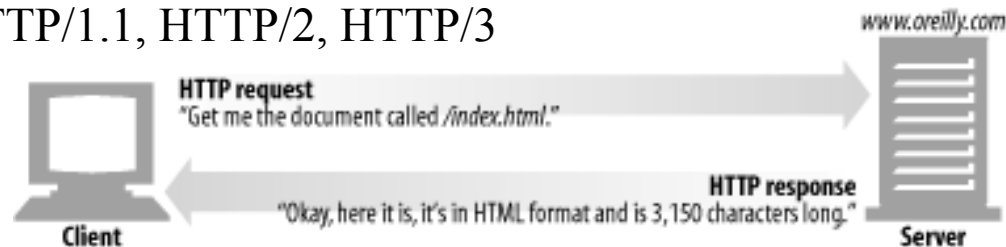
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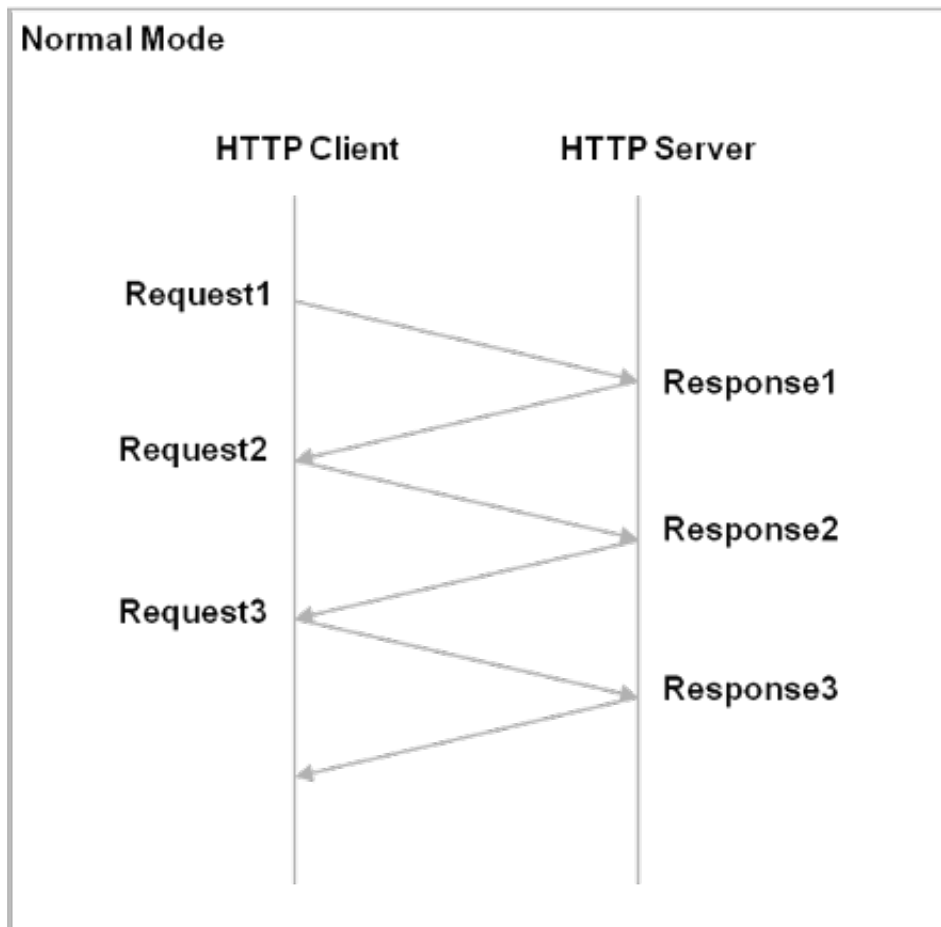
3. HTTP PROTOCOL

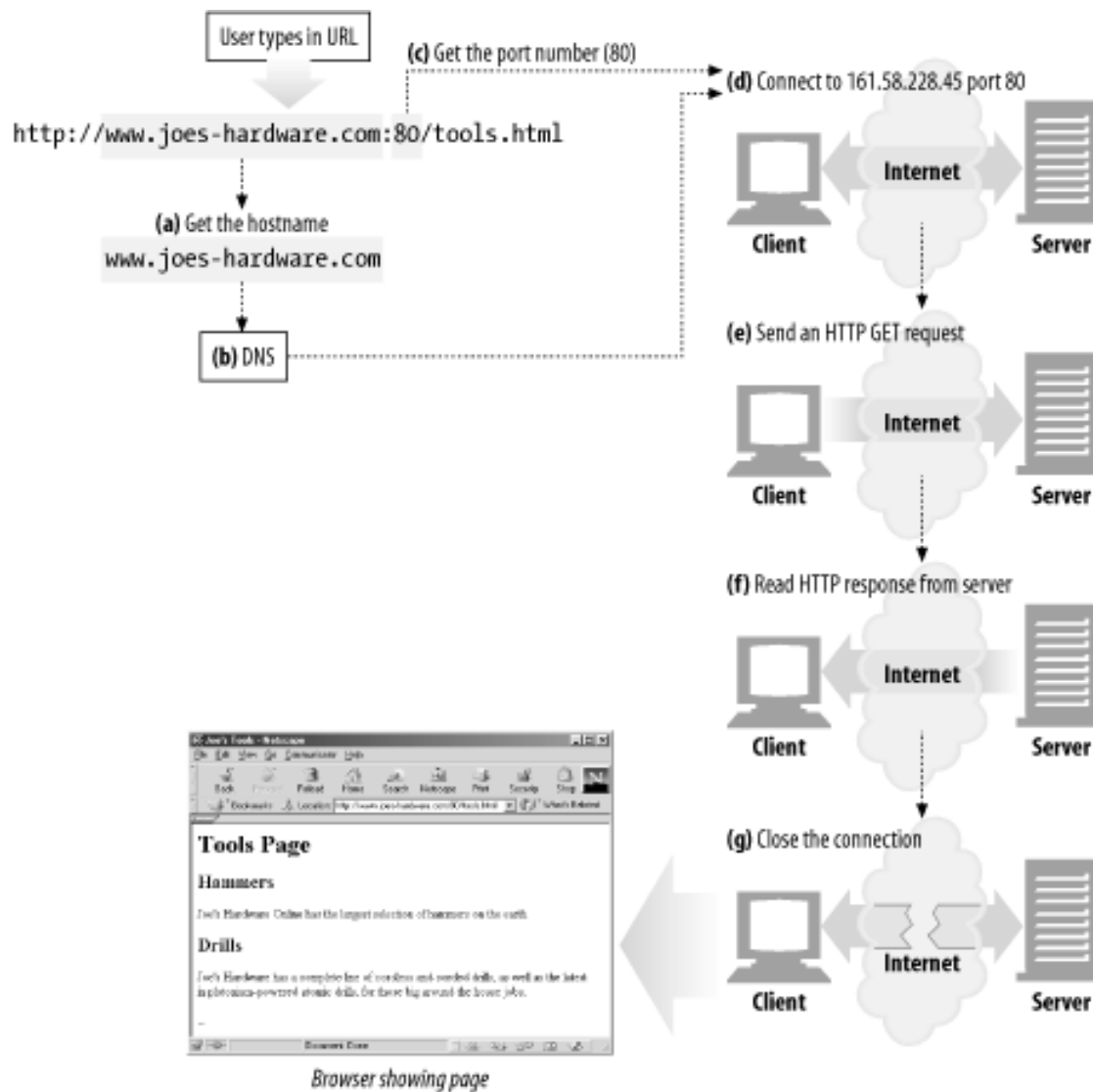
HyperText Transfer Protocol

- Used for communication between a web browser and a web server
- An application layer protocol in the OSI model, operating on top of the TCP/IP protocol
- There are two types of messages: request (web browser), response (web server).
- HTTP servers operate on port 80 by default
- It is a stateless protocol
- Can be used to transmit any type of data
- Versions: HTTP/0.9, HTTP/1.0, HTTP/1.1, HTTP/2, HTTP/3



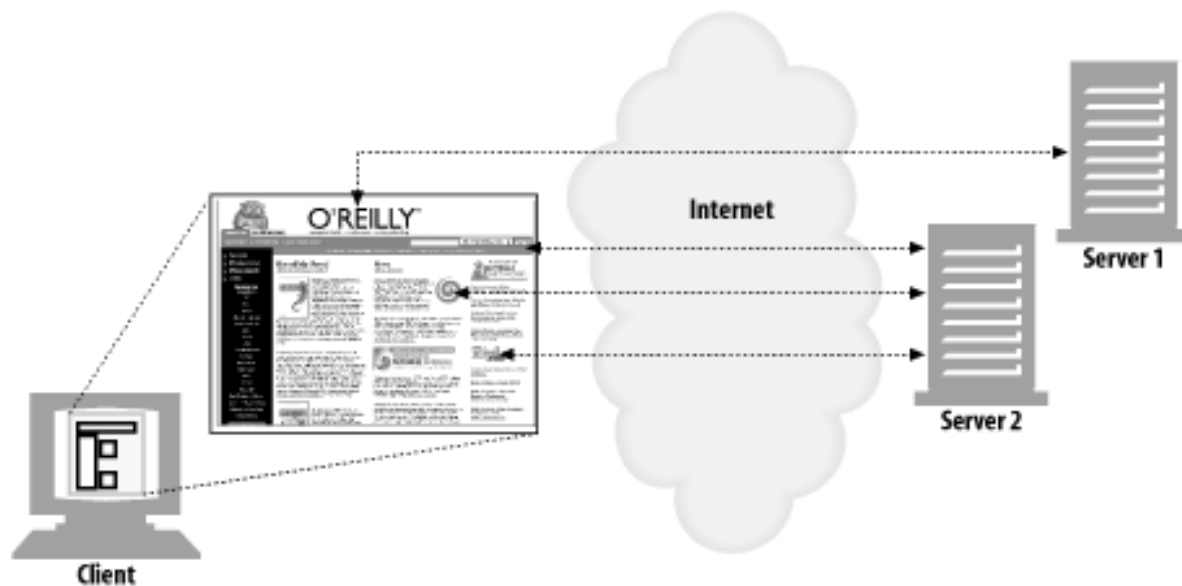
HTTP MODEL

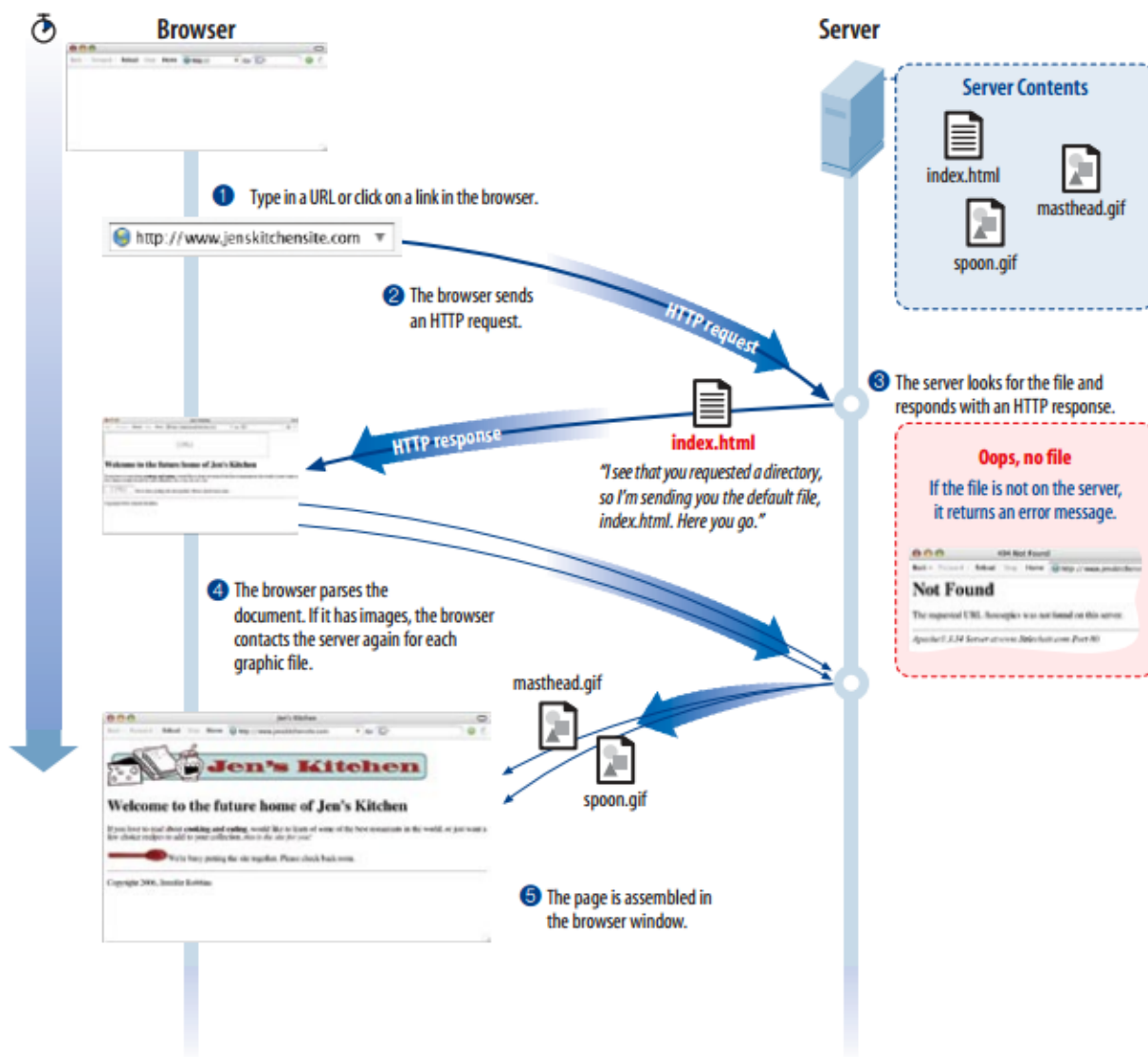




◆ HTTP MODEL

Web pages often contain more than one object => requesting a web page typically generates a series of HTTP messages for each object on the page.





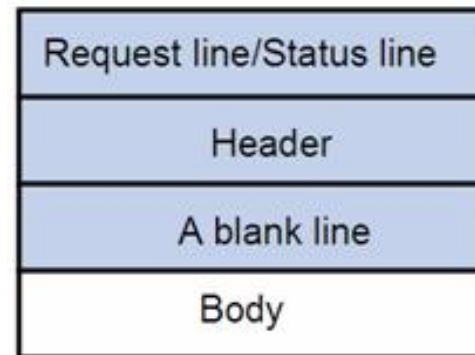
◆ HTTP Message

Plain text, including:

- **Request line/Status line (Start line):** This is the first line, containing the client's request command or the server's response status code.
- **Header:** This may be absent or have multiple lines, each with the following format:

<name>: <value>

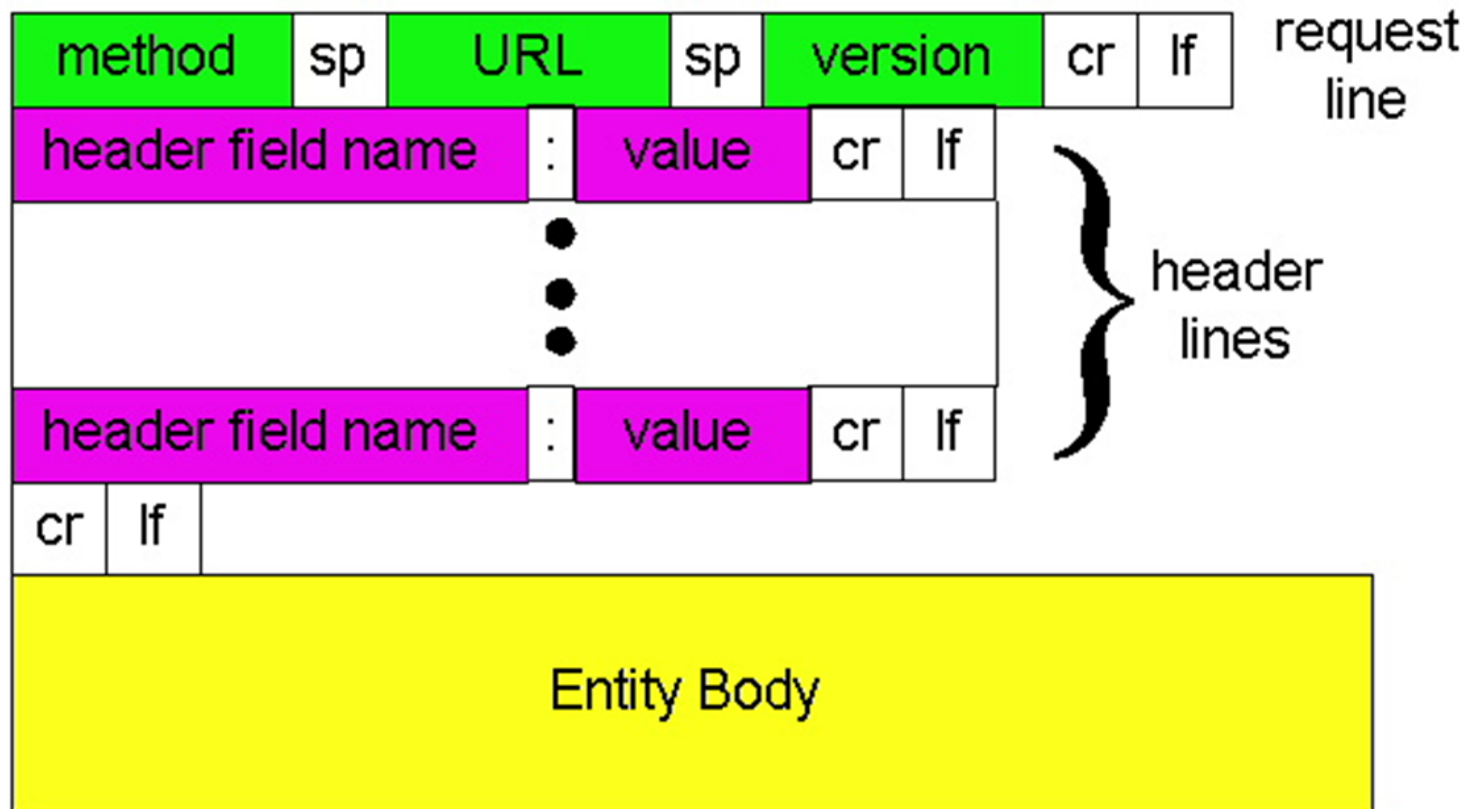
- **A blank line:** a blank line used to distinguish between the header and the body
- **Body:** may or may not be present, and can contain any type of data (plain text, binary)



** Header: for more information:*

http://www.tutorialspoint.com/http/http_header_fields.htm

HTTP Request Message



◆ HTTP Request Message

In which:

- **Method:** the command that the client wants the server to execute
- **URL:** resource address
- **Version:** the HTTP version that the message uses
- **Cr (carriage return) and lf (line feed):** return to the beginning of the line and move to the next line, marking the end of the message.

Method	Description
GET	Retrieve the document specified in the URL
HEAD	Similar to GET, but the return message doesn't have a body.
POST	Provide information to the server
PUT	Upload the document to the server and place it in the location specified in the URL
DELETE	Delete the document located at the URL on the server.

◆ GET vs POST

	GET	POST
BACK button/Reload	Harmless	Data will be re-submitted (the browser should alert the user that the data are about to be re-submitted)
Bookmarked	Can be bookmarked	Cannot be bookmarked
Cached	Can be cached	Not cached
Encoding type	application/x-www-form-urlencoded	application/x-www-form-urlencoded or multipart/form-data. Use multipart encoding for binary data
History	Parameters remain in browser history	Parameters are not saved in browser history

◆ GET vs POST

	GET	POST
Restrictions on data length	Yes, when sending data, the GET method adds the data to the URL; and the length of a URL is limited (maximum URL length is 2048 characters)	No restrictions
Restrictions on data type	Only ASCII characters allowed	No restrictions. Binary data is also allowed
Security	GET is less secure compared to POST because data sent is part of the URL Never use GET when sending passwords or other sensitive information!	POST is a little safer than GET because the parameters are not stored in browser history or in web server logs
Visibility	Data is visible to everyone in the URL	Data is not displayed in the URL

◆ HTTP Request Message

request line
(GET, POST,
HEAD commands)

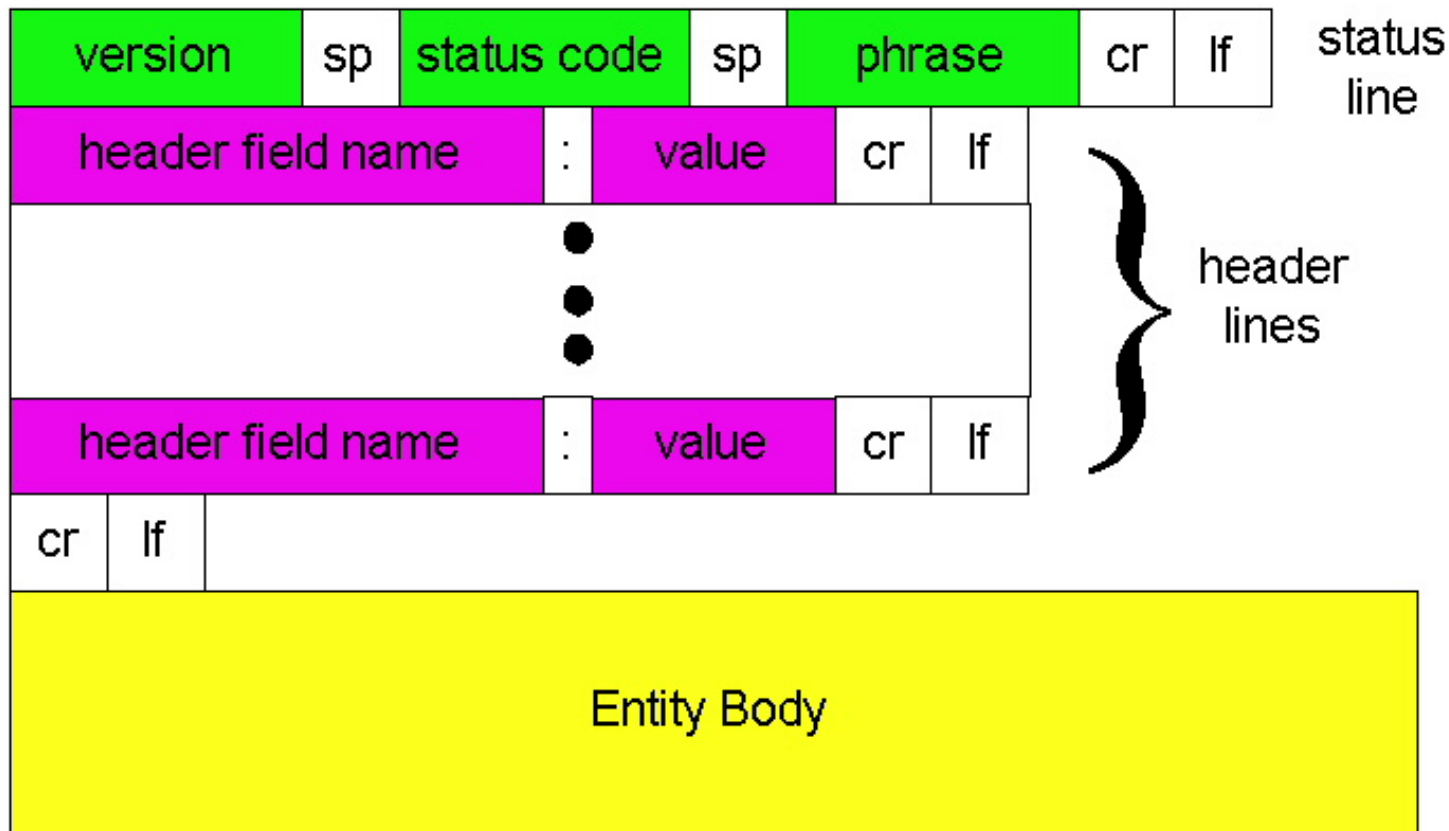
header
lines

```
GET /somedir/page.html HTTP/1.0
User-agent: Mozilla/4.0
Accept: text/html, image/gif, image/jpeg
Accept-language: fr
```

(extra carriage return, line feed)

Carriage return,
line feed
indicates end
of message

HTTP Response Message



◆ Status codes

Code	Type	Reason
1xx	Information	request has been received and continue processing
2xx	Success	request was successfully received, understood, and processed
3xx	Redirection	further action is required to complete the request
4xx	Client error	request contains an error or cannot be fulfilled
5xx	Server error	server failed to fulfill a valid request

◆ Status code

HTTP status code	Description
200	OK. Document returned correctly.
302	Redirect. Go someplace else to get the resource.
404	Not Found. Can't find this resource.

More status codes: http://www.tutorialspoint.com/http/http_status_codes.htm

◆ HTTP Response Message

status line
(protocol
status code
status phrase)

header
lines

data, e.g.,
requested
html file

HTTP/1.0 200 OK

Date: Thu, 06 Aug 1998 12:00:15 GMT

Server: Apache/1.3.0 (Unix)

Last-Modified: Mon, 22 Jun 1998 ...

Content-Length: 6821

Content-Type: text/html

data data data data data ...



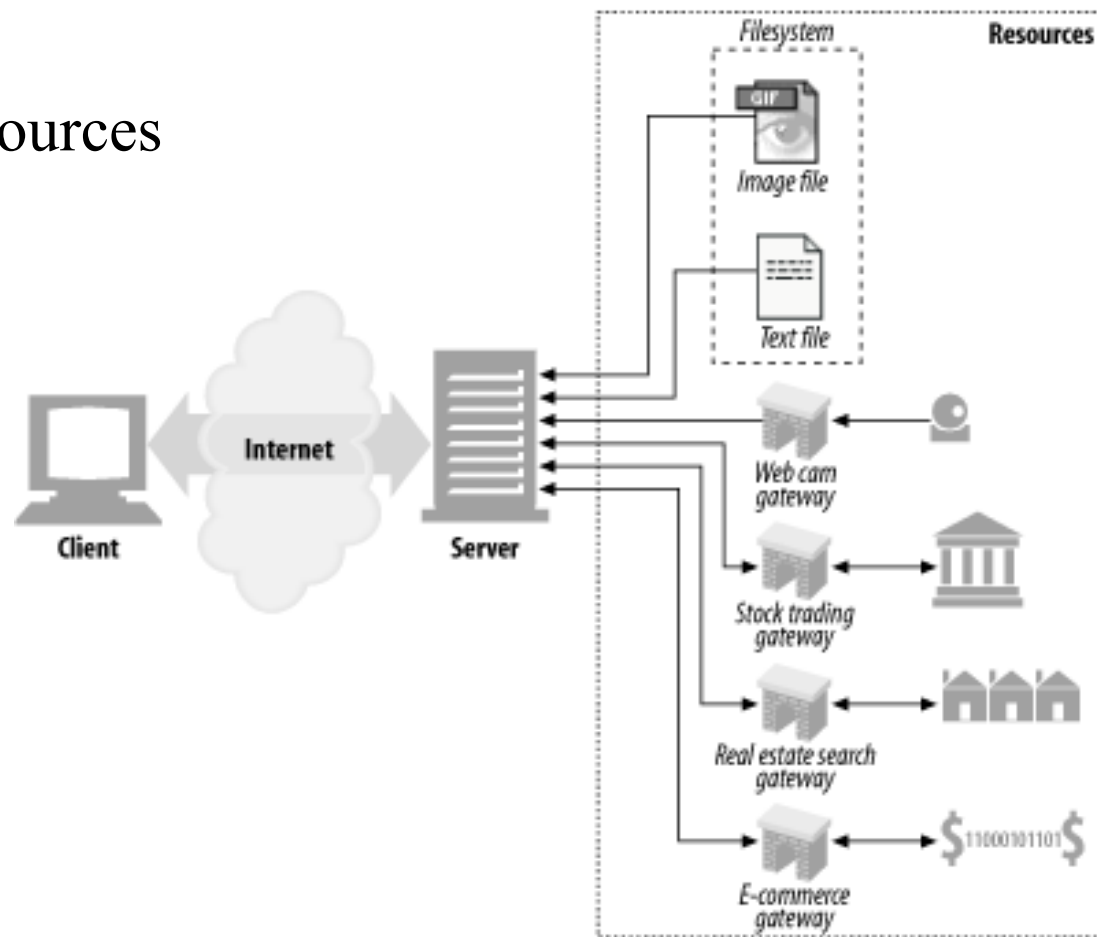


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4. URL

Web resources



URL

URL (Uniform Resource Locator) A resource locator is an address used to locate resources on the Internet.

General syntax:

<protocol>://<user>:<password>@hostpath

In which:

protocol: http, file, ftp, mailto,...

host: the name of the server providing the service Web, FTP,...

path: a local path pointing to a resource on the server

params: include name/value pairs used to provide any additional information needed to access the resource.

query: used to send parameters to the application on the server, the common format is : **name₁=value₁[&name₂=value₂][&...]**

◆ URL



◆ Example URL

HTTP:

`http://<host>:<port>/<path>?<query>`

Example:

`http://www.microsoft.com`

`http://www.joes-hardware.com/inventory-check.cgi?item=12731`

FTP:

`ftp://<user>:<password>@<host>:<port>/<path>;<params>`

Example:

`ftp://ftp.cit.ctu.edu.vn/giaotrinh/`

`ftp://prep.ai.mit.edu/pub/gnu;type=d`

`ftp://ttinternet:ttinternet@172.18.211.19/thuctap/file1.txt`

◆ Example URL

Email:

`mailto:email_address`

Example:

`mailto:ttngoan@cit.ctu.edu.vn`

File on disk:

`file://<host>/<path>`

Example:

`file://internet_server/course/index.htm`

`file:///c:/course/test.html`

◆ Absolute URL & Relative URL

Absolute URL:

is the full address of a resource.

includes the protocol, hostname, path, and filename.

Example: *http://www.cit.ctu.edu.vn/student/index.html*

Relative URL:

is not the full address of a resource.

specifies the resource location relative to the current document's URL

Example:

The current viewing webpage:

http://www.cit.ctu.edu.vn/student/homepage.php

Relative URL = *chuyenmuc.php*

Missing information *http://www.cit.ctu.edu.vn/student/*

Browser automatically determines absolute URL:

URL= *http://www.cit.ctu.edu.vn/student/chuyenmuc.php*

◆ Absolute URL & Relative URL

Relative URL:

Example:

The current viewing webpage:

`http://www.cit.ctu.edu.vn/student/homepage.php`

Relative URL= `../index.php`

Missing information `http://www.cit.ctu.edu.vn/student/`

Browser automatically determines absolute URL:

URL= <http://www.cit.ctu.edu.vn/index.php>

Notation:

`"/` root directory.

`../` the parent directory

◆ URL Encoding

- URLs can only be sent using ASCII character-set
- URLs contain characters outside the ASCII set must be converted into a valid ASCII format => URL Encoding
- URL Encoding: replaces unsafe ASCII characters with a "%" followed by two hexadecimal digits ("*" => %2A)
- URLs cannot contain spaces => replaces with a "+" or %20

VD: `http://www.example.com/new%20pricing.html`

URL Encoding Reference:

http://w3schools.com/tags/ref_urlencode.asp



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◆ 5. HYPERLINK & HYPERTEXT

HyperText: a system that links information elements through triggerable text links, also known as hyperlinks.

HyperLink: a connection between one information element and another. Elements can be:

- Text, hypertext, websites

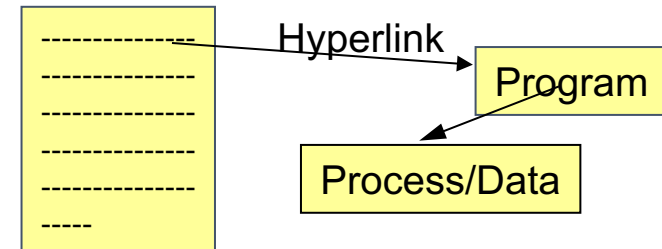
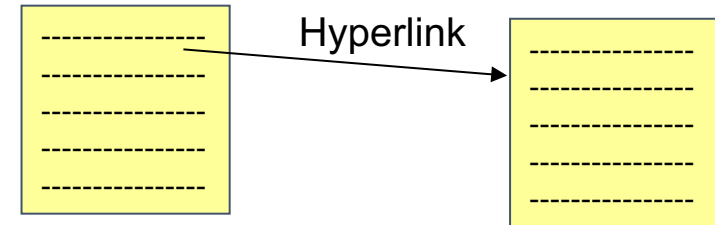
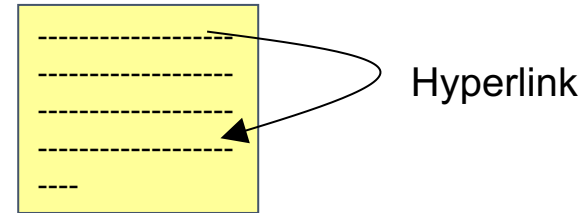
- Audio, images

- Files, ActiveX objects (Word, Excel, etc.)

- Executable programs written in languages such as Java, Java Applet, ASP, ASP.NET, PHP, etc.

◆ Hyperlink types

- Internal link: A link within a document that points to a piece of information within that document itself.
- External link: A link to a different document outside of the document being searched.
- Executable link: An external link that executes a program to process data at the user's request on the Web, and produces resulting information.





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◆ 6. WEB CACHE

Web cache: a mechanism for temporarily storing copies of web resources (HTML pages, images, etc.) to provide to clients without needing to access the original server.

Benefits:

- Reduces client response time because it doesn't require accessing the original server

- Reduces the load on the web server

- Reduces network traffic, saving bandwidth => limits network bottlenecks.

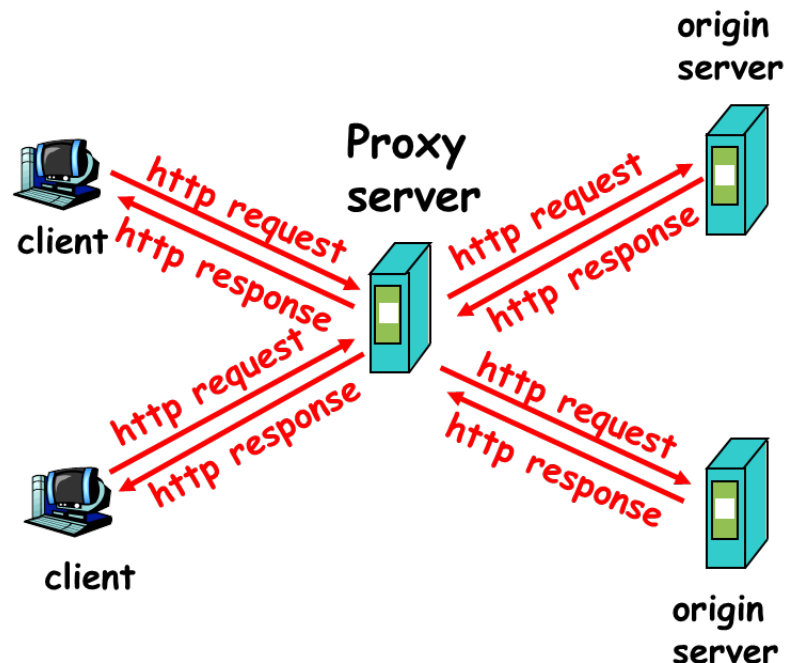
Category:

- Private cache (Browser cache), Public cache (Proxy cache).

Web cache

All client HTTP requests are sent to the web cache.

- If the request object is already stored in the web cache, the web cache will immediately respond to the client.
- Otherwise, the web cache will access the origin server, receive the response, forward it to the client, and also store a copy in the web cache.



Control Web caching

HTML

```
<meta http-equiv="Cache-Control" content="no-cache, no-store, must-revalidate">
```

PHP

```
header("Cache-Control: no-store, no-cache, must-revalidate, max-age=0");
```



Questions?